



COMFOCUS

Community on Food Consumer Science



Guideline for heart rate measures



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1. Guideline for heart rate measures

1.1 Introduction

This guideline presents harmonized protocols for studies that use cardioactive activity as an implicit measure in food consumer science (FCS) context. We recommend the researcher to first and foremost consult COMFOCUS *General guideline for psychophysiological measures* for an overview of common factors to be considered in design and implementation of a study using psychophysiological measures.

This guideline discusses previous applications of electrodermal activity measures in food consumer science (section 1.2), protocol- (section 1.3) and technology- (section 1.4), related factors that should be considered and reported in the study. We also provide recommendations on harmonised measures for HR (section 1.5) and a brief overview on stimuli used in FCS studies that include heart rate measures (section 1.6). For convenience, in section 1.7 we provide a minimum reporting checklist that contains the factors that should be considered and reported in a study using heart rate measures.

1.2 Heart rate and food consumer science

As one of the implicit measures of the autonomic nervous system (ANS) activity, *heart rate* (HR) serves as a valuable real-time indicator of physical and cognitive effort, emotional states, arousal, and attention (Bell et al., 2018b; Kreibig, 2010). Changes in the sympathetic or parasympathetic nervous system as a response to a stimulus result in acceleration or deceleration of the heart rate, usually measured in *beats per minute* (bpm). Another common indicator of cardiovascular activity is a *heart rate variability* (HRV). It measures the interval between successive heartbeats and is considered to be a significant indicator of emotional regulation (Appelhans & Luecken, 2006). In food consumer science, the two most used methods to record such cardiovascular activity are *electrocardiography* (ECG) and *photo plethysmography* (PPG). Generally, ECG is considered a *gold standard* due to a high level of detail and precision in its measures (Laborde et al., 2017). ECG is performed by using electrodes attached to the various parts of the body, typically, chest area, wrists, fingers, ankles. PPG measures pulse rate (a surrogate measure for HR) by shining a LED light onto a skin area where capillaries are easily accessible; the light is then reflected to a sensor that captures blood volume change and translates it into pulse rate. PPG can be obtained via wearables (wrist-, arm-, ear-, temple-, finger-worn devices) or via video recordings (this method called *remote photoplethysmography* (rPPG)). With each cardiac cycle the heart pumps blood to the periphery. Even though this pressure pulse is somewhat damped by the time it reaches the skin, it is enough to distend the arteries and arterioles in the subcutaneous tissue. In remote PPG (rPPG), the change in blood volume caused by the pressure pulse when the face is properly illuminated is detected. The amount of light reflected is then measured. When reflectance is plotted against time, each cardiac cycle appears as a peak. This information can be converted to heart rate average and

variability. Other methods for HR collection include chest straps and pulse oximeters, although they have a higher degree of error compared to ECG (Wang et al., 2017).

One of the principal questions in food consumer science deals with the **affective food experiences**. Since consumer decision making is highly influenced by the emotional response to the products, numerous methods have been used to disentangle the mechanisms behind consumer decision making. With the introduction of psychophysiological measures, researchers have been able to enrich the insights collected via self-reported measures with the more objective – implicit – psychophysiological measures. Heart rate allows to capture a more automatic affective response and emotional processing of the stimuli presented, as has been shown in the studies **comparing implicit and explicit measures** (Kaneko et al., 2021; Walsh et al., 2017). Heart rate has been used in research on **affective reaction to visual cues** (Takada et al., 2018), **hedonic and sensory ratings** (de Wijk et al., 2019a; Torrico et al., 2018), **effect of sensory factors on emotional experience** (Croy et al., 2013; Kantono et al., 2019) **or physiological signals** (Dong & Jacob, 2016), **effect of familiarity or type of stimulus on sensory acceptability and physiological responses** (de Wijk et al., 2021; Sayão et al., 2021a; Torrico et al., 2019), **effect of pathogen cues on affective processing of emotion-eliciting stimuli** (Perone et al., 2021), and **expectation-experience gap effect on the product judgement** (Verastegui-Tena et al., 2019b).

Heart rate measures have also been employed in studies investigating **approach and withdrawal motivation tendencies** (Lagast et al., 2020b) as well as **food neophobia** (Raudenbush & Capiola, 2012). Heart rate has also been used as an indicator of **stress** in works analysing **its impact on food choices** (Vitt et al., 2021) **or behaviours** (Klatzkin et al., 2019; Masih et al., 2019). Some studies also used heart rate as a proxy for **sustained attention** (Soussignan et al., 2015) and **vigilance** (Allen et al., 2014).

Conversely, heart rate variability has been primarily used in studies on **maladaptive eating behaviours**. Since self-regulation is one of the factors in such behaviours, studies have used this measure to investigate **emotional eating** (Juarascio et al., 2020; Maier & Hare, 2017), **inhibitory control and food cravings** (Wu et al., 2020), **implicit wanting** (Kuoppa et al., 2016), **self-regulation as a mechanism behind obesity** (Spitoni et al., 2017), **hedonic reactivity** in face of obesity or eating disorders (Chang et al., 2021; Soussignan et al., 2010), as well as **emotional response to food cues** under eating disorders (Mauler et al., 2006). HRV has also been used in developing and testing new methods to improve self-regulation in eating behaviours and cravings, such **HRV biofeedback** (Meule et al., 2012; Meule & Kübler, 2017).

It should be noted that due to its sensitivity, HR tends to be a difficult measure to interpret; therefore, it tends to be paired with other psychophysiological measures, such as skin conductance (Raudenbush & Capiola, 2012; Soussignan et al., 2010; Verastegui-Tena et al., 2019a), eye tracking (Soussignan et al., 2015), facial expression recognition (de Wijk et al., 2019b; Torrico et al., 2019). Moreover, HR measures are often administered together with self-reported measures. For example, in studies that have employed HR for

measuring affective food experience, participants were also asked to complete the Hedonic liking scale, EmojiGrid, Self Assessment Manikins, Positive and Negative Affect Schedule (Kaneko et al., 2021; Klatzkin et al., 2019; Lagast et al., 2020a; Torrico et al., 2018); for research questions on stress, studies included State Anxiety Questionnaire, Perceived Stress Scale, Trait anxiety, Depression Symptoms (Allen et al., 2014); for studies on maladaptive eating behaviours HR was paired with Food Cravings Questionnaire, three-Factor Eating Questionnaire (Kuoppa et al., 2016; Meule et al., 2012; Meule & Kübler, 2017).

1.3 Protocol

This section describes main factors to consider when designing a protocol for a study involving heart rate measures. For information about the general study design factors and reporting standards please consult *COMFOCUS General Guideline for Measuring Psychophysiological Responses*.

1.3.1 Design

The following table presents heart rate specific factors that should be considered when designing a study. Where applicable, information related to these factors should be reported in the final research output.

Table. Study design factors for studies using heart rate measures

Factor	Recommendations
Study setting	HR measures can be used both in the laboratory and in more naturalistic settings (e.g., home, restaurant, office). Quality of the HR measures highly depends on amount of movement. More static study settings allow researchers to control for quality of signal and movement artefacts since it is possible to use more precise technologies (ECG), provide clear instructions, and supervise the participants. However, use of wearable devices allows studying phenomena in the context that would otherwise be impossible to recreate in the lab. Therefore, the researcher should carefully consider the trade-off that two modes of study settings provide. Also of importance is that heart rate is not instantaneous. If the participant has just arrived at the study setting by mounting a flight of stairs, or has just expressed that they are nervous, then clearly the heart rate needs to settle down to a neutral baseline before the first data is recorded.

Study environment	When using rPPG (i.e., measuring heart rate through colour intensity variation of skin during video recording), an important factor to consider is the lighting. Insufficient light, varying natural light, or changing light source in the room where study is taking place can interfere with the recording. More information on this can be found in Chapter 6 of D6.1 (<i>Guideline for Facial Expression Recognition</i>).
Study timing	If a study is run across multiple days, study design should control for the effect of circadian rhythms by collecting HR measures at the same time of the day (Allen et al., 2014; Kantono et al., 2019; Maier & Hare, 2017). Each session should preferably not be run early in the morning or late in the evening (unless required so by the study design).
Study design	Each study involving HR should include three stages (<i>points in time</i>): baseline (<i>resting</i>), event (<i>reactivity</i>) and post-event (<i>recovery</i>) (Laborde et al., 2017). These stages can be repeated within the study multiple times. It is suggested to use <i>within-subjects</i> design in HRV studies to account for individual differences and decrease the impact of external factors (Quintana & Heathers, 2014). One problem common to the <i>within-subjects</i> design is the habituation to the experimental tasks and a possible learning effect (Quintana & Heathers, 2014). One way to avoid this habituation is to counterbalance the conditions or introduce different tasks that measure the same concept (Laborde et al., 2017).
Sample size	Sample size for studies with HR measures varies greatly between less than 20 participants per condition to more than 100 (Kantono et al., 2019; Kuoppa et al., 2016; Soussignan et al., 2010; Wu et al., 2020). Sufficient sample size should be determined using power calculations. It has been found that HRV studies are generally underpowered, therefore researchers should be precise in setting the effect size and power level prior to calculating the sample size (Quintana & Heathers, 2014).

1.3.2 Sample selection

Participant factors that influence data

Previous research using heart rate measures have underlined a list of potential confounding factors that should be measured and reported (Burmester et al., 2021; Chang et al., 2021; Laborde et al., 2017; Licht et al., 2008; Masih et al., 2019; Meule & Kübler, 2017; Nelson et al., 2020; Quintana et al., 2016; Sayão et al., 2021b; Soussignan et al., 2010, 2015). The list of factors is presented in the Table below. A participant demographic or screening questionnaire which asks participants to record these items would be recommended, and to include them in your inclusion/exclusion criteria where appropriate.

Table. Participant factors that influence heart rate measures

Factor	Details
Physiological influences	• Weight • Height • Waist-to-hip ratio • Level of obesity • Pregnancy or childbirth within last 12 months • Excessive exercise • <i>For wrist-worn devices:</i> wrist circumference • <i>For optical sensors:</i> skin type, hair follicle density, skin tone (including tattoos)¹
Clinical	• Heart conditions (such as arrhythmia) • Cardiovascular diseases • Metabolic conditions (e.g., diabetes) • Renal diseases • Neurological disorders

¹ There is some evidence that a use of optical HR monitors on darker skin tones and tattooed skin might produce less precise results (Nelson et al., 2020). Some research suggests, that in addition to skin tone, sweat and hair follicle density can also have influence on data quality when using wearable devices (Cosoli et al., 2020).

Use of medication	• Use of cardioactive medications • Use of oral contraceptics • Use of psychotropic medications
Stimulants consumption	• Smoking • Substance abuse or dependence • Habitual level of alcohol consumption • Consumption of coffee
Behavioural factors	Certain actions undertaken by the participants before the study can have an unwanted influence on the heart rate measure. It is important to inform the participants of these factors and instruct them on what to do prior to the participation. These usually include (Laborde et al., 2017; Maier & Hare, 2017; Spitoni et al., 2017; Takada et al., 2018): • Following a regular sleep schedule and recording typical bedtime and waking time • Avoiding intense physical activity <i>a day</i> before the study • Avoiding eating meals for 2 hours before the study • No products containing chocolate or cocoa 6h prior to study • Avoid drinking coffee, tea, and caffeinated drinks for 2 hours before the study • Avoid drinking alcohol 24 hours before the study • Use the bathroom before the study (to avoid interruptions) • No medication that alters blood flow (e.g., analgesics) 72 hours before the study

Participant factors that influence safety

Generally, heart rate monitoring is a relatively safe method. However, it should be noted that certain participant characteristics (e.g., clinical conditions like cardiovascular diseases or use of cardioactive medications) have a strong impact on data. However, the researcher cannot ask participants of the study to perform tasks that go against medical recommendations (e.g., engage in physical activity when you have cardiovascular disease) or interrupt regular medication schedule for the participation in the study. This can

raise some ethical issues, in particular related to privacy (sharing medical history) and justice (excluding/discriminating based on medical history).

Inclusion/exclusion criteria

Inclusion and exclusion criteria for HR studies depend highly on the research question. Factors mentioned in this section can all be used as inclusion or exclusion criteria.

Suggested exclusion criteria for studies involving HR measures:

- Clinical factors (*heart conditions (such as arrhythmia), cardiovascular diseases, diabetes, neurological disorders*)
 - *Do you have any pre-existing blood pressure conditions?*
 - *Do you have any mental illnesses, such as severe depression or an anxiety disorder?*
 - *Do you have any persistent heart or respiratory issues?*
 - *Do you have any other medical conditions?*
- Use of medication (*cardioactive medications, oral contraceptics, psychotropic drugs*)
 - *Are you currently taking any medication?*
 - *Are you taking any cardioactive medication?*
 - *If yes, please list the name/s of the medication/s.*
 - *(For female participants) Are you taking any form of oral contraception?*
- Substance abuse or dependence (*alcohol, drugs*)
 - *Have you had any alcoholic drinks in the last 24 hours?*
 - *Do you typically smoke? If yes, please report the number of cigarettes you consume daily.*
 - *Have you smoked recently, within the last two hours?*

Suggested criteria to log and control for, but not exclude:

- Physiological factors (Weight, height, waist-to-hip ratio, level of obesity, pregnancy, or childbirth within last 12 months, skin type, hair follicle density, skin tone (and tattoos))
- Stimulants consumption:
 - *Have you had any caffeinated/theine-containing beverages in the past couple of hours?*
- Physical activity:
 - *Did you rush to arrive to the study place?*
 - *Did you take part in any strenuous physical activity in the past 24 hours?*

- *If yes, what was the activity and its length?*
- *When was the last time you worked out?*
- Food consumption
 - *Have you eaten anything in the last two hours?*
- Sleep schedule
 - *Did you stick to your usual sleeping schedule last night?*
 - *When did you get up this morning?*
 - *When did you go to bed last night?*
- Pre-study questionnaire
 - *Do you need to use the restroom? (to avoid participants moving during the study)*

1.3.3 Ethical factors

HR recordings are biometric data that fall under “special categories of personal data” in the EU General Data Protection Regulation², therefore appropriate requirements for data collection and storage need to be considered prior to the study. See Chapter 2, section **Error! Reference source not found.** of D6.1 for more details on ethical issues across psychophysiological measures. HR data collection will have issues of informed consent (explain risks and procedure, data management), privacy (health data collected as part of screening), potential diagnostics (heart rate measures can be used clinically), bias and accessibility/openness. For more information, please consult COMFOCUS WP8 deliverable *Literature Review on Ethical, Legal and Social Issues (ELSI) relevant to COMFOCUS: A conceptual framework for Responsible Research and Innovation (RRI)*.

1.4 Technological factors

In this section we describe the main technological factors to consider when designing and running a study involving heart rate measures. For a general overview applicable to all psychophysiological studies, please see section 1.2.4 in *General guideline for psychophysiological measures*. The following table presents factors specific to heart rate studies. We recommend first and foremost to follow user manuals supplied by the respective technology providers as the information provided in this section might not reflect the most up-to-date technological possibilities.

² General Data Protection Regulation GDPR, Regulation (EU) 2016/679 (EU)2016/679, 59, European Parliament and of the council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing directive 95/46. 1-88 (Official Journal of the European Union 2018 27 April 2016).

Table. Technological factors that influence heart rate measures

Factor	Details
Hardware	Since data quality and type varies great across different HR data collection methods, it is essential to provide a detailed report on the one used in a study. Generally, ECG provides lower varying degree of error compared to other methods (Nelson et al., 2020). On average, wearable devices (such as wrist-worn monitors) slightly underestimate the absolute HR (between -0.9 and -7.2 <i>bpm</i>) (Nelson et al., 2020). However, they might be more suitable for certain research questions (such as measuring HR in more naturalistic, out-of-lab settings). For long-duration studies using wearable devices, it is recommended to have multiple devices available per participant; if a battery of one device runs out during use, having a substitute will allow for continued observation. Otherwise, charging the device will prolong the overall testing time. Recent developments in wearable devices allow the extraction of HR data from personal consumer devices, such as sports watches; however, their accuracy is generally³ not formally validated against ECG (Quintana et al., 2016). This can cause particular problems when researchers aim to use participants that exhibit above-average error or ectopy (Quintana et al., 2016). The researcher should keep in mind that personal devices might not give access to raw data, provide no details on data correction, and use non-standard (proprietary) metrics (Quintana et al., 2016). Furthermore, the participant should be informed if the commercial website where the data is kept allows the owner of that website to share the data or use it for own purposes. This is often the website of the manufacturer of the sensor device, where the data is kept remotely/in a Cloud.
Sensor type and placement	In general, most heart rate monitors are not adapted to studies involving movement. Excess movement of body areas with sensors will result in so-called <i>motion</i> (movement) artifacts in the data, which happens when the electrodes change their relative position to the skin (Grundlehner & Mihajlović, 2019). Chest-worn monitors can be more adapted to movement; however, they are also more

³ Due to a relatively fast pace of development, The researcher should consult technology providers about such concerns. Some wearable devices have recently gained clearance for standardised ECG features (e.g., Apple Watch 4 and 5 receiving FDA 501(k) class II clearance for the ability to detect arrhythmia (Nelson et al., 2020)).

	invasive to the participants, can result in more motion artefacts due to friction with the skin, and require their electrodes to be hydrated in order to ensure proper conductivity (Laborde et al., 2017; Nelson et al., 2020). It should be noted that pacemakers or implantable cardioverter defibrillators can also interfere with the HR measures when using chest-strap monitors. Wrist-worn monitors are another convenient method for studies including movement, although quality of HR measures can suffer during periods of high activity. In some cases, it might be possible to combine Inertial Measurement Units (IMU) that signal activity with HR measures to discover periods of high activity within the dataset. It should be noted that certain factors can influence data quality when using optical sensors. This includes tightness of the device, placement on dominant vs non-dominant hand, degree of movement, sweating, skin type, and hair follicle density. The researcher should control for these factors in the analysis. When a study is run in the laboratory and therefore the movement is limited, HR sensors should be placed in a way that minimizes the appearance of motion artifacts to ensure high quality of data. This includes placing electrodes away from the (used) limbs when using ECG; putting the wearable device such as wrist-worn pulse oximeter on a non-dominant hand, making sure it is not too loose (nor too tight).
Position	The position (angle) of the body can influence HRV; therefore, the researcher should specify the position of the participants (seated, standing, lying) when measures were collected (Quintana et al., 2016).
Respiration	Some evidence suggests that respiration can provide information related to the cardiac cycle as well as have an influence on high frequency HRV, although others argue that the respiratory and HR oscillations have the same origin, and that monitoring respiration is not necessary (Quintana et al., 2016). When possible, study design should control for different breathing patterns of certain populations, such as athletes, and to monitor for erratic sinus arrhythmia, sudden fluctuations in respiratory rate, and sighs, which could affect HRV. Additionally, HRV is also dependent on HR, and a mathematical correction can be used to improve HRV reproducibility.
Sampling rate	When using ECG, low sampling rate can result in a jitter in the estimations, thus altering the outcome considerably (Shaffer & Ginsberg, 2017). A typical sampling

	rate ranges between 250 and 1,000 Hz. Usually, 500 Hz is considered a sufficient threshold for an accurate measure of native fiducial points, although for healthy adults 250 Hz can provide adequate results as well (Quintana et al., 2016). An adequate sampling rate should be selected based on the intended use of the ECG data and the recommendations by the device provider. The researcher should report the native sampling rate of the hardware. If a lower sampling rate is used and R-wave fiducial points are reconstructed, this should be specified in the study output as well.
Baseline	A few factors need to be kept in mind when deciding on the baseline measures in the study. Participants' path to reach the place where the study is carried out (most typically, the laboratory) might include different levels of physical exertion such as using a staircase. For this reason, sufficient time is needed for all study subjects to acclimatize, rest, and normalize their heart rate. Moreover, participation in lab studies can in itself cause higher levels of emotional arousal due to nervousness, fear, or excitement. One way to approach this is to ask participants to take their seats, attach the sensors and leave them to rest in a relaxed position and/or show them relaxing stimuli. This way participants can both rest from physical exertion (if there was any), get accustomed with the lab setting and sensors. For studies involving computers, experimenters can ask participants to sit relaxed, breathe normally, and look at a blank screen (or close eyes). The length of the baseline can vary highly, depending, among other factors, on the nature of the study, the setting, and the study design. It should be noted that at the beginning of the study there is a higher likelihood for participants to move around and adjust themselves while they are trying to get comfortable. To deal with potential movement artifacts, the researcher can set a longer baseline and use only a later part of it for the analysis (for example, recording the baseline for 10min, but using only the last 5min for the comparisons). For HRV studies two common baseline lengths are either 5min or 10 cycles collecting the lowest frequency of interest (at least 250s in standard frequency domain) (Quintana et al., 2016). However, neither of the two suggestions are obligatory.

Time factors of recording	The timing of the HR measure recording and its use for subsequent analysis should be clearly defined and reported. For example, <i>“heart rate was analysed in the period of 5s before to 5s after the onset of stimuli”</i>. For HRV measures, a suggested minimum length of a continuous recording is 60 seconds (Malik et al., 1996). If HR study is carried out over a long period of time, timestamps should include the time of the recording (e.g., when using wearable devices, the researcher should report the total wear-time per day). Thresholds for wear-time (e.g., minimum of 100 min per day) should also be clearly defined.
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1.4.1 Data processing

The table below provides a list of data processing factors to be considered and reported for studies using heart rate measures.

Table. Data processing factors for studies using heart rate

Factor	Details
Data provided by the equipment	Certain devices might not report standard metrics, give no access to raw data, provide no time markers, and offer no technical details on the methods of correction (Quintana & Heathers, 2014). The researcher should report if the variables were constructed from the raw data or obtained directly from the device.
Data acquisition	HR is typically calculated from a digital raw data stream that is in turn derived from the analog data recorded by electronic or optic sensors. The number of samples needed to create HR from the raw data is typically low, thus HR can be treated as a data modality with very limited latency: for most studies, it is treated as real time (available instantaneously). Artifacts in the signals that construct HR are usually treated by averaging over an odd number of samples. This way, an occasional missing sample has little effect on the instantaneous HR and becomes insignificant when averaged over longer periods (<i>bpm</i> , for instance). The device used will typically have a range of recommended sample rates and other settings with recommendations from the manufacturer.

	Depending on the type of study, certain HR values might be treated as unrealistic. The researcher should report what upper and/or lower thresholds were used to determine which data will be used for the analysis. For example, <i>“intervals exceeding absolute threshold of 160 bpm or intervals deviating by more than 20% from the previous intervals were removed”</i>.
Data cleaning	When using HRV in the study, there is a few sources of artefacts in the data. It is essential to account for and, potentially, clean them. For detailed information on HRV artefact cleaning and transformation, we suggest the review on HRV reported in Quintana et al. (2016). Listwise deletion, which is the practice of completely removing participants from studies due to certain proportions of missing data, is known to cause bias in estimation and negatively affects the accuracy of point estimates (Nelson et al., 2020). Simulation studies with longitudinal data suggest that multiple imputation and full-information maximum likelihood are more reliable in producing better parameter estimates than listwise deletion (Nelson et al., 2020). Additionally, dynamic models of HR can be accurately estimated with high proportions of missing data (e.g., more than 70% of data missing per participant) using full-information maximum likelihood estimation (Nelson et al., 2020). We recommend researchers provide a clear description of data cleaning methods.

1.5 Harmonized measures

When using heart rate, two main categories of HR measures that can be extractes from the data are *heart rate* (number of times heart beats per minute, denoted as squares in the figure below) and *heart rate variability* (the time variation between two consecutive heartbeats, seen as distance between two squares in Figure below). More information on the suggested harmonized measures for studies using heart rate are presented in below.

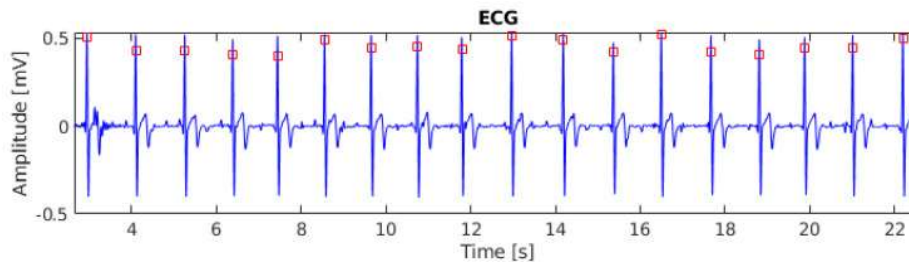


Figure. Example of the ECG signals with annotated heartbeats shown as squares. Adapted from Sieciński et al., (2020).

Table. Harmonized measures for studies using heart rate measures

Factor	Details
Heart rate (<i>bpm</i>)	One of the main cardiovascular measure categories is <i>heart rate in beats per minute</i>. Although it is a relatively straightforward measure of cardiovascular activity, researchers should be aware that it does not give information on the beat-to-beat variation, that describes dynamics of the cardiovascular system. When using heart rate in beats per minute, researchers most often record and compare minimum (HR min) and maximum HR (HR max) levels as well as time in seconds where HR was at maximum (Tmax). Another method is to calculate percentage change in HR compared to the baseline.
Heart rate variability	Another widely used cardiovascular measure is <i>heart rate variability</i>. More than 70 variables alone can be extracted from the data on HRV (Laborde et al., 2017). For this reason, it is essential for the researcher to report the variables used, calculation and analysis methods (such as use of time domain, frequency domain, time-frequency domain, and non-linear domain, as well as Lomb–Scargle periodogram and smoothed pseudo Wigner–Ville distribution). Some of the more common ways to analyse HRV in the time domain are Root Mean Square of Successive Differences (RMSSD), Standard Deviation of Normal-to-Normal Intervals (SDNN), and Standard Deviation of the Average Normal-to-Normal Intervals. For frequency-domain methods, measurements of low frequency (LF), high frequency (HF) and very low frequency (VLF) heart beats as well as ration of HF and LF are typically used. Moreover, HRV metrics depend on the measurement length (24h, short term (~5 min), and ultra-short term (<5min)). Therefore, the

	precise measure will depend on the research question and study design. For more information on HRV metrics and norms, please see Shaffer & Ginsberg (2017)
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1.6 Stimuli

Choice of stimuli in studies that use heart rate measures is highly dependent on research question at hand. Previous studies in FCS have often used **visual** (see e.g., Kaneko et al., 2021; Kuoppa et al., 2016; Meule et al., 2012; Meule & Kübler, 2017; Perone et al., 2021; Sayão et al., 2021b; Soussignan et al., 2015; Spitoni et al., 2017; Takada et al., 2018) and **gustatory** stimuli (see e.g., de Wijk et al., 2019b; Kantono et al., 2019; Lagast et al., 2020; Torrico et al., 2019; Verastegui-Tena et al., 2019). Visual stimuli most often included **pictures** or, less commonly, **videos** (Chang et al., 2021; Walsh et al., 2017) presented on computer screens. Nevertheless, some studies also used **real products** to capture a more complete sensory profile of stimulus (de Wijk et al., 2019b) or other **sensory properties**, such as smell (Croy et al., 2013). Some studies employed wearable devices in more naturalistic, out-of-lab environments (Vitt et al., 2021). In such situations, it might be complicated to report and describe stimuli, therefore the researcher should aim to be as transparent as possible in describing the procedure, hypotheses, and variables of interest.

We recommend the researcher follows the suggested general checklist for stimuli and stimuli-related task description (see **Error! Reference source not found.** in Chapter 2 of D6.1.

1.7 Minimum reporting checklist for studies using heart rate (HR)

Category	Example
Study design	
Study timing	<i>Was study run across several days; if yes, at what time of the day were HR measures collected</i>
Body position	<i>Participants were seated in a reclining chair; participants were standing</i>
Environment	<i>What environment was the study carried out in; if a study was set in the lab, please describe the general environment, including: illumination; sounds; temperature; participant set-up</i>
Timestamps	<i>Are the timestamps for HR measures analysis reported in the study</i>

Baseline	<i>How long was the baseline; which interval of the baseline was used for the analysis; what instructions were given to the participants regarding the baseline; were baseline measures collected multiple times during the study? If so, when and for how long?</i>
Sample	
Participant Factors	<i>Skin tone; BMI; (for wrist-worn devices) wrist circumference, wrist placement (dominant or non-dominant; tight or loose); medical conditions; medication use</i>
Pre-study questionnaire	<i>Information about physical activity (e.g., did the participant rush to arrive to the study place; any physical activity in X hours before study; if yes, activity and length; when was the last time participant worked out)</i> <i>Food/drink consumption (e.g., any food 2 h before the study; any coffee/caffeinated/theine-containing beverages X h before the study)</i> <i>Alcohol consumption (e.g., did participant consumer alcohol in 24h before the study)</i> <i>Tobacco consumption (e.g., is participant a smoker; if yes, what is the number of cigarettes daily; did the participant smoke within the last two hours)</i> <i>Medicine consumption (e.g., any current medications taken; any cardioactive medications; name of them; oral contraception)</i> <i>Medical conditions (e.g., any pre-existing blood pressure conditions, any hear or respiratory issues, other medical condition; any mental illnesses such as severe depression or anxiety disorder)</i> <i>Sleep schedule (e.g., when did the participant go to sleep a night before)</i> <i>Requests before study start (e.g., if participants need to use the bathroom)</i>
Documentation	

Participant Instructions	<i>What were the requirements for the participants before the study (e.g., no coffee 2h before, no intense physical activity a day before the study, etc.)</i>
Technological factors	
Device Make & Model, Serial number	<i>Method; wearable device (how many); technology manufacturer; technology specific details (device type and version, firmware version, sensor); how reliable is the device; justification for choosing such device; (for optical sensors)) what sensor type is used in the device (infrared LED, green LED; both)</i>
Setup	<i>Sensor placement for ECG; on which hand was the wearable device worn</i>
Sensor Type and Placement	<i>Sensor was placed on non-dominant hand; left ribs; wrist-worn monitor on right hand</i>
Respiration	<i>Was respiration measured; was respiration controlled in the study</i>
Sampling rate	<i>Native sampling rate of the hardware; any reconstruction performed</i>
Data	
Data	<i>Raw Data recorded; Processed data (& how processed); What format; Any signal thresholds</i>
Variables	<i>Names of all variables (dependent, secondary, covariates)</i>
Variable Calculation	<i>Definitions (and calculations) for each variable; summary metrics; if multiple samples are taken per minute, final bpm calculations; any second order metric calculation</i>
Cleaning	<i>Artifact identification, data cleaning, data loss (reasons, such as device failure, attrition, battery problems); what upper and/or lower thresholds used for data filtering, if any; how missing data was dealt with; outlier identification and correction</i>

1.8 References

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